



Problems with EKF

- Last week, we studied the EKF as an approach to estimate the state of a nonlinear system.
- EKF is best known and most used nonlinear Kalman filter.
- However, it has serious flaws that can be remedied fairly easily.

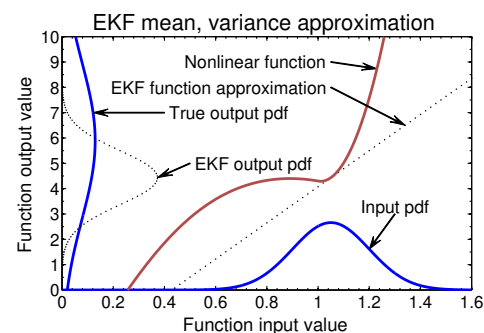
ISSUES: How input mean and covariance are propagated through static nonlinear function to create output mean and covariance estimates.

- When computing mean estimates in Steps 1a and 1c, assumes $\mathbb{E}[f_n(x)] \approx f_n(\mathbb{E}[x])$.
 - This is not true in general, and not necessarily even close to true (depending on “how nonlinear” the function $f_n(\cdot)$ is).
- When finding covariances in Steps 1b and 2a, linearizes Taylor-series expansions.
 - Nonlinear terms are dropped, resulting in a loss of accuracy.



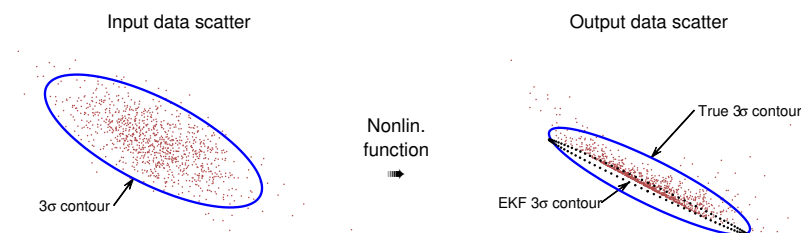
Illustrating problems in 1-d

- A simple 1-d example illustrates these two effects.
- Input RV pdf is shown on horizontal axis, with mean 1.05.
- Inputs passed through red nonlinear function to produce outputs.
- Straight dotted line is linearized approximation used by EKF to find output mean and covariance.
- EKF-approximated pdf is compared to a Gaussian pdf having same mean and variance of the true data on the vertical axis.
- Notice that EKF approach is not producing an accurate estimate of either.



Illustrating problems in 2-d

- Left: cloud of random Gaussian function input points; right: transformed set of output points for 2-d example.
- Actual 3σ confidence interval is compared to EKF-estimated confidence interval.
- “Sigma-point” approach can improve mean, covar propagation through nonlinear equations.





Summary

- EKF is the best-known nonlinear state estimator.
- But, it has some limitations due to assumptions made when approximating expectations.
 - We know that $\mathbb{E}[\text{fn}(x)] \neq \text{fn}(\mathbb{E}[x])$ in general.
 - We know that truncating a Taylor series loses accuracy.
- This week, we study a “sigma point” approach to approximating expectations, results in sigma-point Kalman filter (SPKF/UKF).
- SPKF generally outperforms EKF for nonlinear problems and has same computational complexity as EKF.